EV Battery Course — Module 15: Global and Indian EV Battery Industry

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The global battery industry is concentrated in China: CATL and BYD alone installed 56% of the world's 1,187 GWh of EV batteries in 2025, and China processes 70–95% of key battery minerals. India sold over 2.5 million EVs in FY2025–26 but had only about 1–1.4 GWh of PLI-backed cell capacity running against a 50 GWh target. This module maps the industry and India's position as of October 2026, with every dated figure sourced; it separates operating plants from announced ones.

1. Learning objectives

- Name the major cell makers, their market shares and technology focus.
- Explain mineral supply chains and where concentration risk lies.
- Explain gigafactory economics and the role of utilisation.
- Describe India's EV market, cell-manufacturing projects and policies, and classify each project by status.
- Analyse commercial trade-offs by segment: cars, two-wheelers, three-wheelers, buses, trucks and stationary storage.

2. Global cell makers (2025)

Global EV battery installations were **1,187 GWh in 2025, up 31.7%** from 901 GWh in 2024 (SNE Research, released 4 February 2026).

Rank	Company (country)	2025 GWh	Share	Technology focus
1	CATL (China)	464.7	39.2%	LFP and NMC; CTP (Qilin); sodium-ion
2	BYD (China)	194.8	16.4%	LFP Blade; vertically integrated with its vehicles
3	LG Energy Solution (Korea)	108.8	9.2%	NMC pouch and cylindrical; moving into LFP
4	CALB (China)	62.8	5.3%	NMC and LFP

Rank	Company (country)	2025 GWh	Share	Technology focus
5	Gotion High-tech (China)	53.5	4.5%	LFP; LMFP
6	SK On (Korea)	44.5	3.7%	High-nickel NMC pouch
7	Panasonic (Japan)	44.2	3.7%	NCA/NMC cylindrical (Tesla)
8	Eve Energy (China)	31.3	2.6%	Cylindrical and prismatic
9	Samsung SDI (Korea)	28.9	2.4%	Prismatic NMC; solid-state R&D
10	Svolt (China)	28.5	2.4%	NMC and LFP

3. Mineral supply chains

The IEA's *Global Critical Minerals Outlook 2025* reported that:

- China refines about **70% of lithium chemicals**, processes **70–95%** of global lithium, cobalt, phosphate and graphite, and over **95% of battery-grade graphite**.
- The average share of the top three refining countries for key energy minerals rose from about 82% in 2020 to **86% in 2024**: diversification is not yet happening at refining level.
- Indonesia leads nickel refining, with significant Chinese ownership.

Mineral	Main mining sources	Main refining	Battery use
Lithium	Australia, Chile, Argentina, China, Zimbabwe	China	All Li-ion
Nickel	Indonesia, Philippines, Russia	Indonesia, China	NMC, NCA
Cobalt	DR Congo (dominant)	China	NMC, NCA
Manganese	South Africa, Gabon, Australia	China	NMC, LMFP, LMO
Graphite	China (natural); synthetic in China	China	Anodes
Phosphate	China, Morocco, US	China	LFP

Export controls (e.g. China's graphite controls since 2023, DRC cobalt export quotas in 2025) show that supply can be cut or priced by policy, not just markets.

4. Battery cost trends and gigafactory economics

BNEF's 2025 survey (9 December 2025): average pack price **$108/kWh**, BEV packs **$99/kWh**, LFP packs **$81/kWh**, NMC packs **$128/kWh**, China average **$84/kWh** (see Module 4).

Gigafactory economics. Capex per GWh is much lower in China than elsewhere; materials are about 60–75% of cell cost; yield and utilisation decide profitability. Illustrative example: a 10 GWh plant costing $800 M (an assumption), depreciated over 15 years = $53.3 M/year.

Utilisation	Output	Depreciation per kWh
60%	6 GWh	$8.9/kWh
85%	8.5 GWh	$6.3/kWh
100%	10 GWh	$5.3/kWh

Against LFP cell prices well below $60/kWh in China, a few $/kWh of depreciation from low utilisation is the difference between profit and loss. Global capacity is currently well above demand, which keeps prices low and makes new entrants' ramp-up harder.

5. India's EV market (FY2025–26)

Per JMK Research (7 April 2026):

- Total EV registrations exceeded **25 lakh (2.5 million)**, up **24%** year on year; overall EV penetration about **8.5%** (7.7% in FY2024–25).
- Electric two-wheelers were about **58%** of EV sales; TVS, Bajaj and Hero together held **61%** of the e-2W market.
- Electric cars were the fastest-growing segment (**+86%**). Tata Motors remained the leader, but its share fell from about 57% to about **39%** as JSW MG, Mahindra and Hyundai gained, and VinFast and Maruti Suzuki entered.
- Electric goods vehicles grew **172%** (penetration 1.4%).
- The government's long-term target is **30% EV penetration by 2030**.

6. India's cell-manufacturing projects (status, October 2026)

Company	Location	Capacity planned	Status	Notes
Ola Electric (Ola Cell Technologies)	Krishnagiri, Tamil Nadu	20 GWh PLI allocation; 5 GWh first phase	**Partially operational**: about 1.4 GWh installed (reported June 2025)	4680-format cells
Reliance New Energy	Jamnagar, Gujarat	15 GWh PLI allocation	**Ramping up in 2026** (reported)	Part of an integrated energy complex
Rajesh Exports	Not finalised publicly	5 GWh PLI allocation	**Pre-commissioning**; land acquired	—
Exide Energy	Bengaluru, Karnataka	6 GWh first phase (12 GWh planned overall)	**Commissioning** (July 2026); revenue expected from Q3 FY27	LFP first, then NMC; ~₹4,800 crore invested
Amara Raja (ARE&M)	Telangana	4 GWh first phase; 16 GWh by 2030	**Under construction**; start moved to Q2–Q3 FY27	Cylindrical cells; mostly LFP
Tata Agratas	Sanand, Gujarat	20 GWh (first phase; up to ~40 GWh reported)	**Under construction**	Also building a UK gigafactory

PLI ACC scheme: approved May 2021 with an outlay of **₹18,100 crore** for 50 GWh. As reported to Parliament in March 2026, **40 GWh** has been awarded (Ola 20 GWh, Reliance 15 GWh, Rajesh Exports 5 GWh), land is acquired for all of it, and **only about 1 GWh (Ola) was commissioned**. The original December 2024 target was missed; in late 2025 the government issued show-cause notices and was considering deadline extensions and relaxed localisation norms.

Status key: "commissioning" and "partially operational" mean equipment installed and early production; "under construction" means not yet producing; plans and targets are company statements and frequently slip.

7. Indian policy landscape (as of October 2026)

Policy	What it does	Status
PLI ACC (₹18,100 cr)	Output-linked incentives for 50 GWh of cell manufacturing	Running; delayed
PM E-DRIVE (₹10,900 cr)	Demand incentives and charging infrastructure	Extended to 31 March 2028 for e-buses, e-trucks, e-ambulances; 2W/3W subsidies ended 31 March 2026
Battery component scheme (~₹12,000 cr)	Incentives for cathode, anode, electrolyte, copper foil, separators	Proposed; "final stages of approval" (June 2026)
National Critical Mineral Mission (₹34,300 cr incl. PSU investment)	Exploration, overseas assets (e.g. KABIL lithium blocks in Argentina), processing, recycling, stockpile	Approved January 2025
Critical mineral recycling scheme (₹1,500 cr)	Capex/opex subsidies for recycling Li-ion and e-waste	Approved September 2025
Battery Waste Management Rules 2022	EPR, recovery and recycled-content targets	In force; amended
AIS-038 Rev 2, AIS-156	Battery safety standards for cars and 2W/3W	In force with 2022–23 amendments
PLI Auto	Incentives for advanced automotive technology products including EVs	Running

Industry estimates cited in June 2026 put India's 2030 needs at about 400,000 t of cathode active material and 200,000 t of anode active material for roughly 223 GWh of domestic battery capacity.

8. OEM battery strategies (examples)

OEM	Strategy
Tata Motors	LFP packs; group cell company Agratas for future supply
Mahindra	LFP Blade cells sourced from BYD for INGLO-platform EVs (as reported at launch)
Maruti Suzuki	Entered EVs in FY26; works with Toyota and partners on batteries; has warned about import-dependence risk
Ola Electric	In-house 4680 cells (PLI) for its scooters
TVS, Bajaj, Hero/Ather	Mostly imported cells assembled into packs locally; LFP shift
Global (Tesla, VW, Hyundai)	Mix of in-house cell projects and multi-supplier sourcing; LFP for entry trims

9. Swapping and two-wheeler battery models

Swapping networks (SUN Mobility, Battery Smart, Gogoro India and others) serve e-rickshaws, delivery scooters and fleets, offering battery-as-a-service that lowers vehicle prices. Challenges: lack of standard packs, ownership of battery risk, and capital tied up in spare packs.

10. Segment trade-offs

Segment	Chemistry trend	What matters most	Charging/swapping	Commercial notes
Passenger cars	LFP mass market; NMC premium	Cost per km, range, fast charging, resale	Home AC + DC public	Battery 30–40% of car cost
Two-wheelers	LFP increasingly; some NMC	Price, safety, heat tolerance	Home charging, some swapping	End of PM E-DRIVE 2W subsidy in 2026 tests economics

Segment	Chemistry trend	What matters most	Charging/swapping	Commercial notes
Three-wheelers	LFP; lead-acid in low-end e-rickshaws	TCO, uptime	Swapping and depot	Strong commercial case
Buses	LFP	Life (cycles), safety, depot charging	Depot DC	Supported by PM E-DRIVE through 2028
Trucks	LFP/NMC	Energy, payload, charging time	High-power DC, swapping trials	Early; incentives added
Stationary storage	LFP; sodium-ion emerging	Cost per cycle, safety	Grid	India's installed storage only ~8.7 GWh in mid-2026, but ~260 GWh in development (reported)

11. Engineering and commercial calculations

Market concentration: CATL + BYD = (464.7 + 194.8) ÷ 1,187 = **55.6%** of 2025 EV installations.

India PLI progress: 1.4 GWh installed ÷ 50 GWh target = **2.8%** (mid-2025 data).

Illustrative cell import bill: if India's EV sales in FY2025–26 averaged 3 kWh per 2W (1.45 M units) and 35 kWh per car (about 0.18 M units, assumption), cell demand ≈ 4.4 + 6.3 = **~10.7 GWh**, plus 3W, buses and storage. At $60/kWh, that is roughly **$0.6–1 billion** a year of mostly imported cells. (Assumptions labelled; adjust with actual data.)

12. Common mistakes and misconceptions

- **"Announced GWh = capacity."** India's announced pipeline exceeds 100 GWh, but operating capacity is a few GWh.
- **"Localising pack assembly localises the battery."** Cells and materials are 70–80% of value.
- **"Mining is the bottleneck."** Refining and processing concentration is the bigger risk.
- **"Low cell prices mean easy profits."** They squeeze margins for new entrants.

13. Practical case study: localisation decision for a cell supply contract

Problem. An Indian OEM must choose between a Chinese LFP cell supplier and a new Indian cell maker for a 2028 model.

Analysis. Chinese supply: proven quality and lowest price, but tariffs, geopolitical risk and localisation-incentive penalties. Indian supply: incentives and supply security, but unproven yields, schedule risk and possibly higher price initially.

Decision. Dual-source: launch with the proven supplier; qualify the Indian supplier in parallel with staged volume commitments tied to PPAP, yield and cost milestones; design the pack to accept both cells.

Lesson. Supply-chain strategy is risk management, not only price.

14. Revision notes

- 2025 global EV battery installations: 1,187 GWh; CATL 39%, BYD 16%.
- China dominates refining (70–95%); graphite >95%.
- Pack prices 2025: $108/kWh average; LFP $81/kWh.
- India FY26: 2.5 M+ EVs, 8.5% penetration; cars +86%.
- PLI ACC: 40 of 50 GWh awarded; about 1–1.4 GWh operating.

15. Glossary

Term	Meaning
ACC	Advanced chemistry cell
PLI	Production-linked incentive
CAM / AAM	Cathode / anode active material
Utilisation	Actual output ÷ nameplate capacity
BaaS	Battery-as-a-service
KABIL	Khanij Bidesh India Ltd, overseas mineral acquisition company

16. Knowledge check

Conceptual

1. Why is refining concentration a bigger risk than mining concentration?
2. Why does utilisation matter so much in gigafactory economics?
3. Why has India's PLI ACC fallen behind schedule?
4. Why are LFP and India a natural fit?
5. What does the end of 2W subsidies mean for e-2W makers?

Numerical

1. A plant's fixed costs are $70 M/year. Compute per-kWh fixed cost at 5 GWh and 8 GWh output.
2. If India's EV penetration rises from 8.5% to 30% of 2.6 crore vehicles, how many EVs per year is that?
3. Using 2025 data, what share did the top three Korean makers hold together?

Application

1. Draft a supply-risk register for an Indian 2W maker's battery supply.
2. You are advising on where in the battery value chain India should invest first. Argue for one segment.

17. Answer key

Conceptual

1. Ore can come from many countries, but almost all of it must pass through Chinese refineries to become battery-grade chemicals; a refining disruption hits everyone.
2. Depreciation and fixed costs are spread over output; low utilisation raises cost per kWh when cell margins are thin.
3. Know-how gaps, technology licensing, equipment and materials imports, ramp-up complexity, financing, and aggressive targets.
4. Cost sensitivity, hot climate (thermal stability), no nickel or cobalt (mineral security), and two- and three-wheelers dominate volumes.
5. Price competitiveness must come from cost reduction (cheaper cells, LFP, localisation) and financing rather than subsidy.

Numerical

1. 70 ÷ 5,000,000 kWh = **$14/kWh**; at 8 GWh = **$8.75/kWh**.
2. 0.30 × 26,000,000 = **7.8 million EVs/year** (assuming total vehicle sales of 2.6 crore).

3. LGES 9.2% + SK On 3.7% + Samsung SDI 2.4% = **15.3%**.

Application

1. Risks: single-country cell supply; export controls on graphite/LFP technology; currency; supplier quality; regulatory changes (AIS-156); EPR costs; price volatility of lithium. For each: probability, impact, mitigation (dual sourcing, inventory buffers, index-linked contracts, supplier audits).
2. Example argument: cathode active material (LFP) and precursor processing, because it captures large value, aligns with the component scheme and LFP demand, and reduces exposure to China's refining dominance; alternatively, recycling, given BWMR targets and the 2025 incentive.

18. Further learning

- IEA *Global EV Outlook* (annual) and *Global Critical Minerals Outlook 2025*.
- BNEF Lithium-Ion Battery Price Survey (annual press summary).
- NITI Aayog reports on ACC manufacturing and India's EV transition.
- JMK Research and EVreporter quarterly India EV market reports.
- IEEFA, "Assessing India's incentive scheme to enhance battery manufacturing ecosystem."

Sources

- CnEVPost: Global EV battery market share in 2025 (SNE Research, 4 Feb 2026)
- Mercom India: China's dominance in critical minerals continues: IEA (28 May 2025)
- BNEF: Battery pack prices fall to $108/kWh (9 Dec 2025)
- JMK Research: India's EV penetration reached 8.5% in FY2025-26 (7 Apr 2026)
- pv magazine India: Land acquisition completed for 40 GWh under PLI ACC (25 Mar 2026)
- Outlook Business: Govt plans relaxations as battery push runs into delays (1 Dec 2025)
- Outlook Business: Exide's Bengaluru cell plant to generate revenue from Q3 (10 Jul 2026)
- The Machine Maker: Amara Raja postpones cell plant launch to FY27
- MeraEV: India's battery gigafactory scoreboard (29 Jul 2026)
- Energética India: PM E-DRIVE extended to March 2028 (8 Aug 2025)
- Outlook Business: India's ₹12,000 cr battery component push (18 Jun 2026)
- PM India: National Critical Mineral Mission approved (29 Jan 2025)
- PM India: ₹1,500 cr critical mineral recycling scheme (3 Sep 2025)

Next: Module 16, next-generation battery research.